When Kernel Affine Hull Geometry Helps Prediction: Local Coordinates, Decoder Capacity, and Conditional Gains

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Abstract

Kernel Affine Hull Machines (KAHMs) learn nonlinear local reconstruction geometry, but good reconstruction need not yield a useful predictive code. We separate reconstruction, score encoding, and decoder capacity using matched local banks, controlled target variation, and reproducible forecasting experiments. On a straight-branch law, affine reconstruction is nearly exact yet its matched score code nearly suppresses a predictive reflection distinction; KAHM scores retain that distinction. Locality-aware affine scores and explicit local coordinates remove most of this gap, identifying positional information as the useful property rather than reconstruction accuracy alone. Adding affinity-gated local coordinates improves KAHM ridge prediction in eight of nine primary settings. A separately frozen follow-up with 24 fresh repetitions per forecasting law confirms a conditional KAHM benefit: with 256-landmark nonlinear decoders, KAHM local coordinates reduce mean Van der Pol error by 47.3% relative to raw inputs and 19.1% relative to matched affine local coordinates. Both advantages exceed prespecified practical margins under simultaneous paired intervals. It loses to raw inputs on nonlinear autoregression and to both controls on Duffing. We provide a validation-selected implementation, numerical-stability analysis, and complete experiment scripts. The findings support task-dependent KAHM representations, while leaving a transferable criterion for their advantage open.

1 Introduction

Kernel Affine Hull Machines (KAHMs) provide a way to learn nonlinear reconstruction maps from local reference sets. Their original motivation includes geometrically informed classification and privacy–utility tradeoffs (Kumar et al., 2024). A natural next question is whether this geometry helps prediction beyond the tasks for which it was introduced. That question needs more precision than a comparison of reconstruction errors or average ranks on a collection of datasets. A reconstruction map, the numerical code passed to a predictor, and the predictor itself are different objects. An advantage at one stage need not survive the next.

Consider a local model that reconstructs every point on an affine line exactly. Its residual is zero everywhere on that line. The reconstruction is excellent, but the residual does not specify where a query lies along the line. If the target varies along that direction, a predictor receiving only the residual cannot use the information that the reconstructor itself used. Conversely, a less accurate nonlinear reconstruction can induce scores that vary with local position and are therefore useful to the prediction task. This observation does not establish a general preference for inaccurate reconstruction. It identifies a missing link: the code must preserve the variation relevant to the target, and the selected decoder must be able to use that variation.

We investigate that link for KAHM representations. Our experimental question is: *which property of KAHM geometry helps prediction, under what conditions, and how should that knowledge change the model?* We address it at two levels. A controlled branch distribution holds the input law fixed while changing whether the target depends on branch identity or position along the branch. Matched geometric banks isolate the reconstruction rule from reference selection, local principal directions, and score normalization. Forecasting experiments then test whether a practical representation suggested by the controlled analysis has predictive value under finite decoder budgets.

The investigation gives a conditional positive result for KAHMs. On the straight-branch law, the KAHM score code preserves a reflection distinction that the matched affine score code nearly removes. This matters when the target depends on signed position. It does not arise from lower input reconstruction error. Adding local positional coordinates repairs the affine code as well, and changing the affine scoring rule also substantially reduces the gap. Thus the result identifies a useful property of the KAHM code in this comparison, rather than a property exclusive to KAHMs.

The forecasting result is different and complementary. KAHM affinities with a linear decoder are poor on the two oscillator laws. Retaining local coordinates improves those representations, and a nonlinear decoder gives a further substantial improvement on Van der Pol. We used the first-stage observation to specify a separate follow-up before generating new data. Across 24 fresh repetitions, KAHM local-coordinate features outperform both raw histories and matched affine local-coordinate features on Van der Pol at each of two landmark budgets. The corresponding comparisons favor other representations on Duffing and nonlinear autoregression. The controlled branch mechanism therefore motivates an architectural change, while the forecasting study establishes its conditional utility without claiming that the branch mechanism completely explains the oscillator result.

The paper makes three contributions.

1. A controlled KAHM investigation separates reconstruction fidelity, information retained by score encoding, and accessibility to a chosen decoder. A fixed-input target intervention and matched reconstruction controls reveal a positional-information mechanism and its limits.
2. A fresh-sample capacity study establishes a reproducible KAHM advantage on a declared forecasting law, with paired comparisons, simultaneous intervals, practical effect thresholds, and unfavorable results on neighboring benchmark laws retained in the same analysis.
3. An executable research artifact implements the resulting model design: retain local position, allow a nonlinear decoder, and select the complete representation–decoder pipeline using validation data. The artifact preserves the historical results and regenerates experiments and paper assets from explicit configurations.

Local tangent features are established constructions (Yu et al., 2009; Yu & Zhang, 2010), and the score-normalization inverse used below is elementary algebra. We do not present them as new learning principles. The contribution is their controlled use to understand KAHMs, the conditional empirical finding, and the tested consequence for KAHM model design. The supporting mathematical statements explain what the experiments can and cannot establish; they are not new KAHM approximation rates or distribution-free performance certificates.

2 Related work

KAHMs and prediction from geometric reconstruction. KAHMs were introduced for geometrically informed classification, including a differential-privacy application (Kumar et al., 2024), and developed for collaborative learning (Kumar et al., 2025). We reuse their local reconstruction operators through the pinned source of Kumar (2026). Geometric classification has earlier antecedents: nearest-feature-line classification measures distance to affine lines through same-class examples (Li & Lu, 1999), and kernel sparse representation studies reconstruction-based coding in an implicit feature space (Gao et al., 2010). Such results do not establish that scalar reconstruction affinities retain variation needed for another predictive target. We examine that transition while matching partitions, references, local bases, scores, and decoders. This evaluates a KAHM component's predictive role, not the original methods' classification or collaborative-learning claims.

Local coordinate coding and local prediction functions. Local coordinate coding connects localized representations to linear approximation of smooth functions (Yu et al., 2009). The closest construction is local tangent coding: Yu & Zhang (2010) augment coding weights by products with displacement along local principal directions. Our feature form $[\phi_c(z),\ \phi_c(z)P_c^\top(z-\mu_c)]_c$ instantiates this established augmentation.

With a linear readout it is also a soft combination of local affine functions, related to mixtures of experts (Jacobs et al., 1991). Here the gates are fitted without outcomes and frozen during supervised decoding. We ask whether KAHM-derived gates improve this architecture relative to matched affine-derived gates. We neither introduce local tangent features nor transfer their approximation bounds without verifying their locality and smoothness assumptions.

Target relevance and information accessible to a decoder. Supervised dimension reduction seeks features that preserve the relationship between inputs and responses (Fukumizu et al., 2004). Our branch intervention changes the target while keeping input samples, reconstruction bank, signal variance, and noise variance fixed. Predictive $\mathcal{V}$-information provides a complementary distinction: informativeness depends on an observer's model family, with restricted regression as one special case (Xu et al., 2020). We similarly distinguish retained information from its accessibility to a specified fitting procedure. Ridge and finite-feature RBF comparisons do not estimate optimal risk over all measurable decoders; an augmentation gain can reflect accessibility or regularization as well as information recovery.

Normalization and coordinate choice. Retaining the normalizer of positive affinities restores their unnormalized values in exact arithmetic. Our log-normalizer experiment applies this elementary identity to KAHM folding scores. Log-partition information has established uses, including energy-based out-of-distribution detection (Liu et al., 2020), although that work concerns classifier logits and a different task. We compare score coordinates with an invertible reparameterization: information agrees while finite-decoder performance can differ. The outer-affinity normalizer is distinct from the signed-reference denominator inside KAHM reconstruction. We claim neither the inverse identity nor log-normalizer augmentation as a new general principle.

Finite kernel approximation and model selection. Nyström approximation uses a smaller landmark system (Williams & Seeger, 2001), whose size can affect statistical regularization and computation (Rudi et al., 2015). Our capacity study therefore tests two declared procedures, not equivalent function classes or an unrestricted raw-input kernel predictor. Model selection can also induce optimistic evaluation when the selection criterion is reused to assess performance (Cawley & Talbot, 2010). We distinguish validation selection, follow-up on independent synthetic data, and numerical sensitivity on reused observations. These distinctions concern the validity and scope of the comparison; they do not introduce a new selection or inference method.

Relation to the earlier KAHM prediction-limit studies. Kumar & Kargaran (2026) study successor soft coordinates, class-centered lower certificates for norm-bounded linear predictors, and Koopman closure. A related unpublished manuscript (Anonymous, 2026) develops a risk-certificate hierarchy for observed-output forecasting under declared decoder sensitivity. Those works already distinguish fitted accuracy from class risk and use KAHM features, independent evaluation, and geometric comparisons. We do not claim those ingredients or extend the certificate hierarchy. Our question concerns how reconstruction, encoding, and retained local position interact with the target and finite decoder. The fixed-input intervention, encoding ablations, and separately frozen capacity follow-up address this question. Their paired-repetition intervals are not distribution-free class-exclusion certificates, and the supporting propositions supply no new KAHM approximation rate.

3 Separating Geometry, Encoding, and Prediction

3.1 A shared local bank

Let $x \in \mathbb{R}^d$ denote an input and let z be its training-standardized version. A bank contains three blocks with 2, 4, and 8 local models. Within each block, training-only k-means defines the reference groups. Each local model c uses at most 24 shared reference observations, their mean μ_c, and a shared orthonormal basis P_c obtained from the pinned KAHM implementation. The effective rank is at most two, at most $d-1$, and at most the number of local references minus one. For two-dimensional branch inputs the rank is one. Across the

two main geometric alternatives, we keep the scaler, partitions, reference indices and values, basis, effective rank, score definition, and association rule fixed.

The affine reconstruction is

$$r_c^{\mathrm{A}}(z) = \mu_c + P_c P_c^\top (z - \mu_c). \tag{1}$$

The KAHM reconstruction uses the upstream regularized kernel construction. Writing the local reference matrix as V_c, the query-kernel row as $k_c(z)$, and the fitted regularized matrix as B_c, its algebra is

$$a_c(z) = \frac{k_c(z)B_c}{k_c(z)B_c\mathbf{1}}, \qquad r_c^{\mathrm{K}}(z) = V_c^\top a_c(z)^\top. \tag{2}$$

The weights can be signed; this is not a convex combination guarantee. The source implementation uses float32 reference observations and replaces an exactly zero denominator by one. We retain both conventions in the main experiments and study the zero-denominator issue separately in Section 6.4. The repository vendors the unchanged upstream autoencoder and kernel routines at an immutable source commit.

This design compares reconstruction rules within a common local bank. It does not compare separately optimized, complete KAHM and affine learning systems. A centroid reconstruction and alternative native scoring formulas provide secondary controls. Here "native score" means the corresponding score formula evaluated on the shared references, not an independently tuned native pipeline.

3.2 Scores and representations

For either reconstruction, define the radial and angular components

$$q_c(z) = 1 - \exp\{-\|z - r_c(z)\|_2\}, \tag{3}$$

$$a_c^{\mathrm{ang}}(z) = \pi^{-1} \arccos\left(\mathrm{clip}\left(\frac{\langle z, r_c(z)\rangle}{\|z\|_2\|r_c(z)\|_2}, -1, 1\right)\right), \tag{4}$$

$$T_c(z) = \frac{\sqrt{q_c(z)^2 + a_c^{\mathrm{ang}}(z)^2}}{\sqrt{2}}. \tag{5}$$

When the angular denominator is zero, the cosine is set to zero, following the implementation. For each block, the affinity is

$$w_c(z) = (1 - T_c(z) + \tau)^\omega, \qquad Z(z) = \sum_{j \text{ in block}} w_j(z), \qquad \phi_c(z) = w_c(z)/Z(z), \tag{6}$$

with $\omega = 16$ and $\tau = 10^{-6}$. A block therefore produces relative scores, rather than preserving their absolute scale.

We study several representations formed by concatenating across blocks: the affinities ϕ; the affinities with one $\log Z$ per block; the full score vector T; the affinities with standardized raw inputs; and the local-coordinate representation

$$\Gamma_{\mathrm{local}}(z) = \left[\phi(z),\ \left\{\phi_c(z)P_c^\top(z - \mu_c)\right\}_{c=1}^{14}\right]. \tag{7}$$

The latter uses the query's position in each local basis, not the reconstructed vector. It has the familiar form of gated local tangent coordinates (Yu & Zhang, 2010). Our question is how KAHM-derived gates behave within this construction. The same augmentation is applied to affine gates. Forecast inputs have dimension 12, score codes dimension 14, $[\phi, \log Z]$ dimension 17, and local-coordinate features dimension 42. These stored dimensions are not assertions of equal effective capacity.

There is a useful sufficient condition for complete information retention. Since every ϕ_c is positive in exact arithmetic, dividing each gated coordinate by its gate recovers $P_c^\top(z - \mu_c)$. If the matrix formed by stacking all $P_c^\top$ has full column rank, these projections determine z and the local code is injective (Proposition 3). Under that condition, performance differences between raw inputs and local codes concern finite fitting and decoding, including regularization and numerical conditioning, rather than Bayes information loss. A

retrospective audit finds full numerical column rank in all 144 forecasting banks, with positive gates on the inspected test queries (Appendix C.4). These checks support the relevance of the sufficient condition; they do not constitute a symbolic rank proof or guarantee numerically stable inversion. Rank deficiency alone does not establish information loss, because the gates may retain additional information beyond the projections.

3.3 Information preservation does not imply equal decoding difficulty

The outer normalization in Equation (6) has a simple exact inverse if its normalizer is retained:

$$T_c = 1 + \tau - \exp\left\{ \frac{\log \phi_c + \log Z}{\omega} \right\}. \tag{8}$$

Thus $(\phi, \log Z)$ and T contain the same information in exact arithmetic. They need not give the same fitted risk under a linear decoder, a finite nonlinear basis, feature standardization, or regularization. In particular, an improvement from ϕ to $[\phi, \log Z]$ cannot, by itself, identify whether information recovery or easier decoding caused the gain. The full-score control helps distinguish these possibilities. This outer normalizer is also distinct from the signed kernel denominator in Equation (2).

For scalar prediction, write $m(x) = \mathbb{E}[Y \mid X = x]$. The Bayes error using a representation $U = \Gamma(X)$ satisfies

$$R_U^{\mathrm{Bayes}} - R_X^{\mathrm{Bayes}} = \mathbb{E}\left[(m(X) - \mathbb{E}[m(X) \mid U])^2 \right]. \tag{9}$$

A fitted predictor can have additional approximation, regularization, estimation, and selection error. We use flexible heads to probe accessibility, but finite-head experiments cannot identify the Bayes error in Equation (9). Claims about a fitted pipeline are kept separate from claims about information loss.

4 A Controlled Test of Target-Relevant Geometry

4.1 Change the target while keeping the inputs fixed

Let S be uniform on $\{-1, +1\}$, let T be uniform on $[-1, 1]$, and let ϵ be uniform on $[-0.1, 0.1]$, independently. The controlled law is

$$X = (T, c\cos(\pi T) + 0.5S), \qquad Y_\lambda = \sqrt{1-\lambda}\, S + \sqrt{3\lambda}\, T + \epsilon, \tag{10}$$

where $c \in \{0, 1\}$ and $\lambda \in \{0, 1/2, 1\}$. The input identifies both T and S. The target's signal variance equals one and its Bayes MSE equals $1/300$ in every setting. At $\lambda = 0$ the signal is branch identity; at $\lambda = 1$ it is signed position; at $\lambda = 1/2$ both matter.

Within a repetition and curvature, all three targets use the same input samples, noise samples, and unsupervised fitted bank. Across curvatures, the latent samples are also shared. Consequently, varying λ changes predictive relevance without changing input geometry or signal variance. It can still change the difficulty of representing the conditional mean within a particular decoder family. We therefore analyze both fixed-head contrasts and additional representation controls.

4.2 A reflection diagnostic and its mathematical interpretation

The transformation $Q(t, x_2) = (-t, x_2)$ preserves the branch input law. Its target-odd component is

$$m_{\mathrm{odd}}(X) = \{m(X) - m(QX)\}/2 = \sqrt{3\lambda}\, T, \qquad \|m_{\mathrm{odd}}\|_{L^2} = \sqrt{\lambda}. \tag{11}$$

A representation satisfying $\Gamma(QX) = \Gamma(X)$ almost surely cannot recover this component. Its Bayes risk is at least $1/300 + \lambda$. This statement allows the representation to retain even functions such as $|T|$; it does not require the representation to be constant along a branch.

For approximate symmetry, let $\delta_\Gamma^2 = \mathbb{E}[d(\Gamma(X), \Gamma(QX))^2]$ in a specified feature metric. Any κ-Lipschitz decoder obeys the supporting inequality

$$R(g \circ \Gamma) - R_X^{\mathrm{Bayes}} \geq \left(\|m_{\mathrm{odd}}\|_{L^2} - \kappa \delta_\Gamma / 2 \right)_+^2. \tag{12}$$

The proof is an even–odd decomposition and the reverse triangle inequality; we include it in the appendix. The experiment measures reflection discrimination, but does not replace δ_Γ by an empirical estimate and assert a population certificate. The fitted heads also have no prescribed common Lipschitz budget. This bound supplies an interpretation of the controlled intervention, rather than an additional certified empirical result.

4.3 Why reconstruction alone is an incomplete criterion

If m is L-Lipschitz and an input can be reconstructed as $h(U)$ from the *predictor's code* with mean squared error at most $\varepsilon_{\mathrm{rec}}^2$, conditional expectation gives

$$R_U^{\mathrm{Bayes}} - R_X^{\mathrm{Bayes}} \le L^2 \varepsilon_{\mathrm{rec}}^2. \tag{13}$$

This familiar argument requires reconstruction through U. A local reconstructor $r_c(X)$ receives X itself. Small $\|X - r_c(X)\|$ does not establish the existence of h when the predictor receives only $\phi(X)$. Further, even the existence of h does not ensure that a restricted fitted decoder can approximate $m \circ h$ well. The experiment tests precisely this distinction between the geometry's use of an input and the final code's retention of it.

5 Experimental Design

5.1 Data laws and independent evaluation units

The primary first-stage experiment has nine settings: six branch laws at training size 1,024 and three forecasting laws, namely nonlinear autoregression (AR), Van der Pol, and Duffing. Branch validation and test sizes are 512 and 2,048 points. A nested training size of 256 is secondary. Each forecasting repetition generates 128 training, 128 validation, and 512 test episodes. An episode has length 72; histories have length 12; the prediction horizons are 1, 4, and 12; origins have stride four. Each episode supplies 13 origins. All origins from an episode stay in one split. The loss first averages squared errors across origins and horizons within an episode, then across episodes. The oscillator laws are the specified numerical generators, including their integration schemes, rather than claims about exact continuous-time dynamics.

We additionally evaluate the UCI Airfoil Self-Noise data (Brooks et al., 1989) as a secondary static-regression check. Entire operating-condition groups, defined by the four non-frequency inputs, are held out. The fixed split has 64 training, 21 validation, and 21 test groups, containing 905, 295, and 303 rows. Losses weight groups equally. The 24 fitting seeds reuse this same dataset and split. They measure fitting variability, not 24 independent natural-data replications. This dataset supplies a useful adverse result but does not constitute a natural forecasting evaluation.

5.2 Decoders and validation

The primary decoder is ridge regression with an unpenalized intercept. Training-only standardization precedes fitting, and validation selects the regularization parameter from seven values between 10^{-8} and 1. A nonlinear diagnostic uses a Gaussian Nyström basis augmented by the linear input coordinates (Williams & Seeger, 2001). Training rows supply the landmarks. Three bandwidth multipliers and the same ridge grid are selected on validation data; the basis columns are standardized before ridge fitting. Thus this diagnostic is a finite RBF-plus-linear model, not an exact kernel-ridge estimator with its usual RKHS penalty and not an oracle for all possible decoders.

Forecast horizons share validation-selected hyperparameters. Branch targets select independently. Models remain fitted on the training set: validation observations are not incorporated in a subsequent refit. Predictions are clipped to the known support, $[-1, 1]$ for the forecasting generators and $[-2.1, 2.1]$ for the branch law; Airfoil predictions are not clipped. Raw ridge, quadratic ridge, and raw RBF models are included as controls.

5.3 Two stages with distinct comparison families

Two pilot repetitions check execution and set practical effect margins. Before the first confirmation data are generated, an internal hash manifest records source, configurations, analysis specification, pilot records, and margins. This is a recorded internal freeze, not a public prospective registration. The first stage independently regenerates training, validation, and test data in each of 24 repetitions. Common randomness supports paired comparisons within a repetition.

For each of the nine primary settings, define

$$G = R(\phi^{\mathrm{A}}, \mathrm{ridge}) - R(\phi^{\mathrm{K}}, \mathrm{ridge}), \tag{14}$$

$$N = R(\phi^{\mathrm{K}}, \mathrm{ridge}) - R([\phi^{\mathrm{K}}, \log Z^{\mathrm{K}}], \mathrm{ridge}), \tag{15}$$

$$L = R(\phi^{\mathrm{K}}, \mathrm{ridge}) - R(\Gamma^{\mathrm{K}}_{\mathrm{local}}, \mathrm{ridge}). \tag{16}$$

Positive values favor KAHM geometry, normalizer retention, and local-coordinate retention, respectively, in the indicated comparisons. The 27 cellwise contrasts plus $G(0) - G(1)$ at each curvature form a family of 29. We use two-sided paired t intervals across 24 repetitions, with Bonferroni family level 0.05 and 23 degrees of freedom. A cell's practical margin is 5% of the mean pilot raw-ridge MSE. These intervals are parametric approximations, not distribution-free certificates. A non-significant contrast is not an equivalence finding.

The second stage was designed after observing promising first-stage Van der Pol results with KAHM local features and an RBF decoder. Its plan and practical margins were frozen before 24 fresh repetitions per forecasting law were generated from separate random-number streams. It crosses four representations—raw inputs, affine local coordinates, KAHM local coordinates, and KAHM scores—with 64 and 256 landmarks. The 12 primary contrasts compare KAHM local coordinates against raw inputs and affine local coordinates in each law and width. They have their own 95% Bonferroni family of paired t intervals. The practical margin is 5% of that law's first-stage mean raw-RBF MSE, fixed before the new outcomes. We do not assert a joint 95% event across the two families.

Matched landmark seeds and nominal widths constrain the comparison but do not equalize effective capacity: feature dimensions and retained Nyström ranks can differ. Landmark sets at different widths need not be nested. The follow-up tests robustness across two declared finite budgets, not superiority over every possible raw-input predictor. All required repetitions completed; no failed seed was silently removed or replaced.

6 Results

6.1 Useful positional discrimination can coexist with worse reconstruction

Affine reconstruction has lower held-out reconstruction error in every repetition of each of the five main input-geometry settings: straight and curved branches and the three forecasting laws. This holds for both the best-local-model and affinity-weighted reconstruction summaries. The reconstruction advantage does not translate into a uniform predictive advantage of the affine score code.

On the straight branches with the signed-position target, mean ridge MSE is approximately 1.002694 for affine affinities and 0.024668 for KAHM affinities. The primary paired KAHM advantage is 0.978026 with simultaneous interval $[0.96255, 0.99350]$. Yet the average best-local affine reconstruction error is approximately 2.16×10^{-16}, compared with 0.00328 for KAHM. These reconstruction figures are squared Euclidean errors divided by input dimension in standardized input space, averaged over blocks; their role is diagnostic rather than an additional primary test.

The reflection diagnostic explains an important part of this result. Using training-standardized feature coordinates, the RMS difference between $\phi(X)$ and $\phi(QX)$ averages about 7.5×10^{-9} for affine scores and 1.40 for KAHM scores. The affine representation is nearly reflection-invariant in this setting, whereas KAHM scores distinguish opposite signed positions. We do not assume that a finite learned bank has this property in every law: on curved branches both codes substantially distinguish reflected queries. The diagnostic is descriptive and its scaling is not silently substituted into Equation (12).

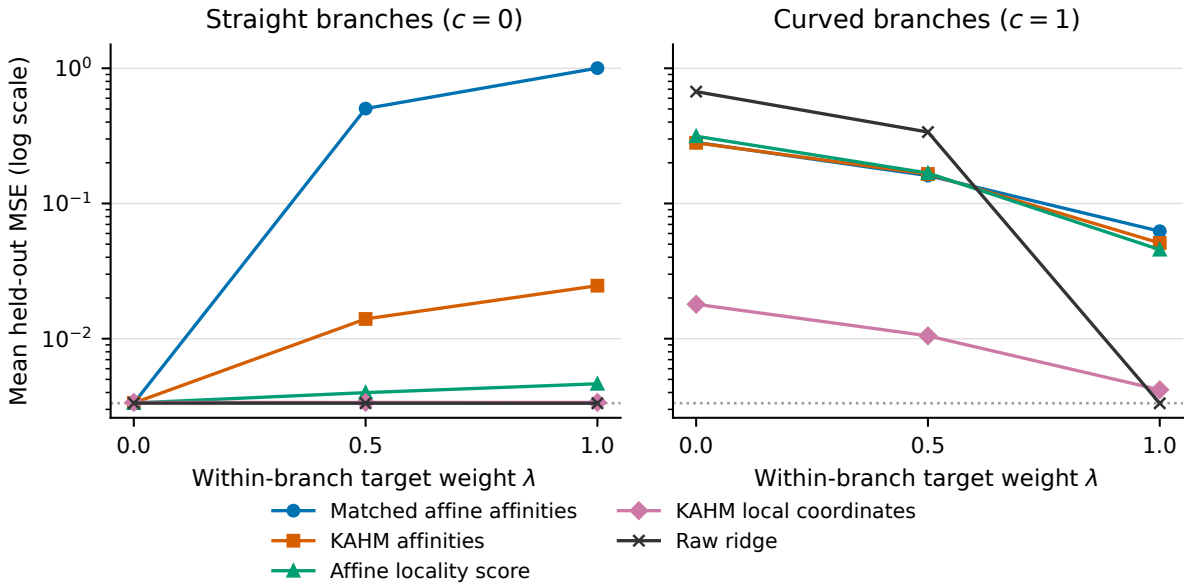


Figure 1: Target relevance changes while branch inputs, fitted banks, and noise samples are held fixed within a repetition. The score-only affine code performs poorly when signed position matters on straight branches. Explicit local coordinates and a locality-aware affine scoring control remove most of the gap. The dotted line is the Bayes MSE 1/300. The displayed mean errors summarize 24 repetitions; only the declared contrasts in the text belong to the primary interval family.

Table 1: The six primary branch settings at 1,024 training points: mean ridge test MSE over 24 repetitions, with the primary KAHM advantage G and its interval from the 29-comparison family. The Bayes MSE is 1/300. The affine-locality scoring control reuses the shared references and is secondary.

c	λ	Affine ϕ	KAHM ϕ	Affine locality	G [interval]
0	0	0.003334	0.003347	0.003344	−0.000013 [−0.000031, 0.000005]
0	0.5	0.503448	0.013995	0.003991	0.489453 [0.481962, 0.496944]
0	1	1.002694	0.024668	0.004641	0.978026 [0.962552, 0.993500]
1	0	0.282887	0.281126	0.314394	0.001761 [−0.071260, 0.074781]
1	0.5	0.160813	0.165207	0.167971	−0.004395 [−0.043672, 0.034882]
1	1	0.062366	0.051042	0.045555	0.011324 [0.000882, 0.021767]

The scoring controls are essential to interpreting the large contrast. A locality-aware affine score gives mean error 0.004641, and affine local coordinates give 0.00333, close to the Bayes error and raw ridge. Therefore, the large advantage over matched affine affinities is not evidence that affine geometry is intrinsically unable to predict this target. It is evidence that a residual-based encoding can discard positional information and that KAHM's induced score variation can retain it. An explicit position-preserving code achieves the same practical purpose more directly.

The primary target interaction also constrains the interpretation. On straight branches, $G(\lambda = 0) - G(\lambda = 1) = -0.978039$, with simultaneous interval $[-0.99351, -0.96257]$. The KAHM advantage is greater when signed position matters, rather than when branch identity alone matters. The corresponding curved-branch interaction is unresolved. These results disfavor a simple account in which the useful KAHM property is merely separation of branch regimes. The native-score controls support the more specific positional interpretation, although those controls were not members of the formal interaction contrast.

Table 2: All nine first-stage primary settings: mean test MSE over 24 repetitions for ridge decoders. The local-coordinate improvement L and its interval belong to the 29-comparison family. Retaining local position substantially improves KAHM prediction in eight settings; KAHM local coordinates do not dominate affine local coordinates.

Setting	KAHM ϕ	KAHM local	Affine local	Raw	L [interval]
$c = 0, \lambda = 0$	0.003347	0.003378	0.003337	0.003338	−0.000031 [−0.000043, −0.000020]
$c = 0, \lambda = 0.5$	0.013995	0.003379	0.003333	0.003334	0.010616 [0.010033, 0.011200]
$c = 0, \lambda = 1$	0.024668	0.003380	0.003333	0.003330	0.021287 [0.020199, 0.022376]
$c = 1, \lambda = 0$	0.281126	0.017982	0.048916	0.671776	0.263144 [0.217187, 0.309101]
$c = 1, \lambda = 0.5$	0.165207	0.010497	0.026756	0.337620	0.154710 [0.122867, 0.186552]
$c = 1, \lambda = 1$	0.051042	0.004188	0.004656	0.003329	0.046854 [0.040400, 0.053307]
Nonlinear AR	0.005300	0.004867	0.004867	0.005053	0.000433 [0.000368, 0.000498]
Van der Pol	0.111108	0.014475	0.011582	0.041340	0.096633 [0.092436, 0.100830]
Duffing	0.053729	0.007560	0.004448	0.006764	0.046168 [0.041729, 0.050608]

6.2 Local coordinates repair a predictive bottleneck, without making it KAHM-specific

Adding gated local coordinates to KAHM affinities improves ridge prediction in 8 of the nine primary settings. In all eight, the simultaneous interval clears the prespecified practical margin. The exception is the straight-branch, identity-only target, where the small adverse difference is below that margin in magnitude. Adding $\log Z$ improves seven settings under the primary intervals, five by more than their practical margins. Neither intervention is a universal improvement.

The three forecasting laws show the scale of the local-coordinate effect (Table 2). On Van der Pol, KAHM ridge MSE falls from 0.111108 to 0.014475; on Duffing it falls from 0.05373 to 0.00756. However, affine local features are still better under ridge on both laws. The first-stage geometry contrasts alone therefore do not establish a KAHM advantage on the oscillators. They establish that score-only KAHM codes are an inadequate predictive interface for these restricted heads.

The inverse in Equation (8) is verified numerically with maximum error about 2.8×10^{-16} on the checked representations. Even so, the first-stage Van der Pol KAHM-score RBF error is 0.00158, whereas the information-equivalent $[\phi, \log Z]$ RBF error is 0.01273. This secondary comparison illustrates that preserving information does not equalize access to it under a particular finite decoder and tuning scheme. It also cautions against interpreting every local-coordinate gain as a proof of irreversible information loss in the original representation.

6.3 Fresh data confirm a conditional KAHM forecasting advantage

Table 3 reports the second-stage comparison. On Van der Pol, KAHM local-coordinate features outperform both alternatives at 64 and 256 landmarks in all 24 repetitions. At 256 landmarks, mean MSE is 0.000871969 for KAHM local coordinates, 0.001078003 for affine local coordinates, and 0.001654150 for raw histories. The corresponding relative reductions, computed from the means, are 19.1% and 47.3%. These percentages are descriptive ratios; the primary inferential quantities are paired absolute differences.

At 256 landmarks, the simultaneous lower endpoints of the raw-minus-KAHM and affine-minus-KAHM intervals are 0.00075496 and 0.00018100, respectively (Table 3). Both exceed the frozen practical margin 0.00014891. The two corresponding 64-landmark intervals also clear that margin. Thus the positive result survives the tested increase in nonlinear decoder budget. Increasing the budget improves raw and affine performance too; this is not evidence of an asymptotic separation between representation families.

The adverse comparisons matter for the scope of the result. On nonlinear AR, raw histories outperform KAHM local features at both widths; the KAHM–affine differences are unresolved. On Duffing, both raw and affine local features outperform KAHM local features at both widths. At 256 landmarks, mean Duffing MSE is about 0.002798 for affine local features and 0.004074 for KAHM local features. All four corresponding

Table 3: Fresh second-stage forecasting results: mean MSE over 24 new repetitions per law, with $M = 64$ or 256 RBF landmarks. All MSEs, differences, and interval endpoints are multiplied by 10^3. The difference Δ is control minus KAHM-local MSE; positive favors KAHM. Intervals belong to the separate 12-comparison family. The same decoder family and tuning grid are used, with gains on Van der Pol and losses on the other laws.

System	M	Control	Control MSE	KAHM local MSE	Δ [interval]
Nonlinear AR	64	Raw	4.4238	4.8795	−0.4556 [−0.5235, −0.3878]
Nonlinear AR	64	Affine local	4.9120	4.8795	0.0325 [−0.0031, 0.0680]
Nonlinear AR	256	Raw	4.6958	5.1749	−0.4791 [−0.6013, −0.3569]
Nonlinear AR	256	Affine local	5.2347	5.1749	0.0598 [−0.0131, 0.1327]
Van der Pol	64	Raw	2.8586	1.0266	1.8320 [1.7369, 1.9271]
Van der Pol	64	Affine local	1.3767	1.0266	0.3501 [0.2755, 0.4247]
Van der Pol	256	Raw	1.6541	0.8720	0.7822 [0.7550, 0.8094]
Van der Pol	256	Affine local	1.0780	0.8720	0.2060 [0.1810, 0.2311]
Duffing	64	Raw	3.4597	4.2154	−0.7557 [−1.0831, −0.4282]
Duffing	64	Affine local	3.0549	4.2154	−1.1605 [−1.4707, −0.8503]
Duffing	256	Raw	2.8364	4.0736	−1.2372 [−1.5242, −0.9503]
Duffing	256	Affine local	2.7981	4.0736	−1.2756 [−1.5702, −0.9810]

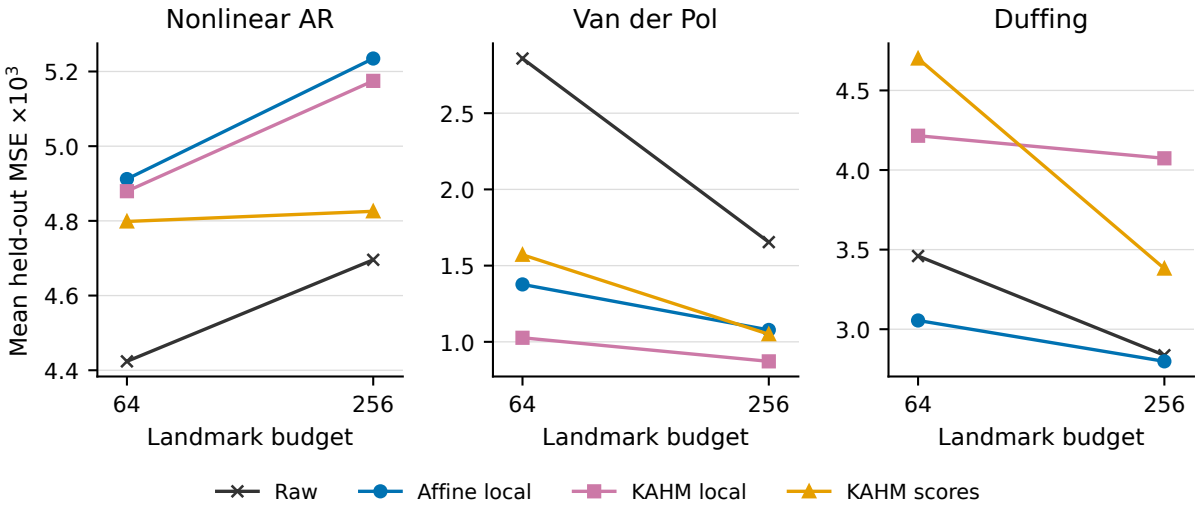


Figure 2: Mean second-stage forecasting errors as the nominal nonlinear decoder budget increases. Feature dimensions and effective Nyström ranks differ, so the horizontal axis is a landmark budget rather than an exactly equalized model capacity. The ordering depends on the forecasting law.

adverse raw/affine comparisons have simultaneous intervals excluding zero. The advantage therefore does not follow from the label "nonlinear oscillator" alone.

These results establish the utility of the declared KAHM pipeline on the specified Van der Pol law and sampling design. They do not establish a task-independent geometric statistic that predicts the advantage. In particular, the straight-branch reflection experiment does not prove that the same symmetry explains Van der Pol. Connecting an independently measured geometric diagnostic to a held-out task's KAHM advantage is a remaining research problem.

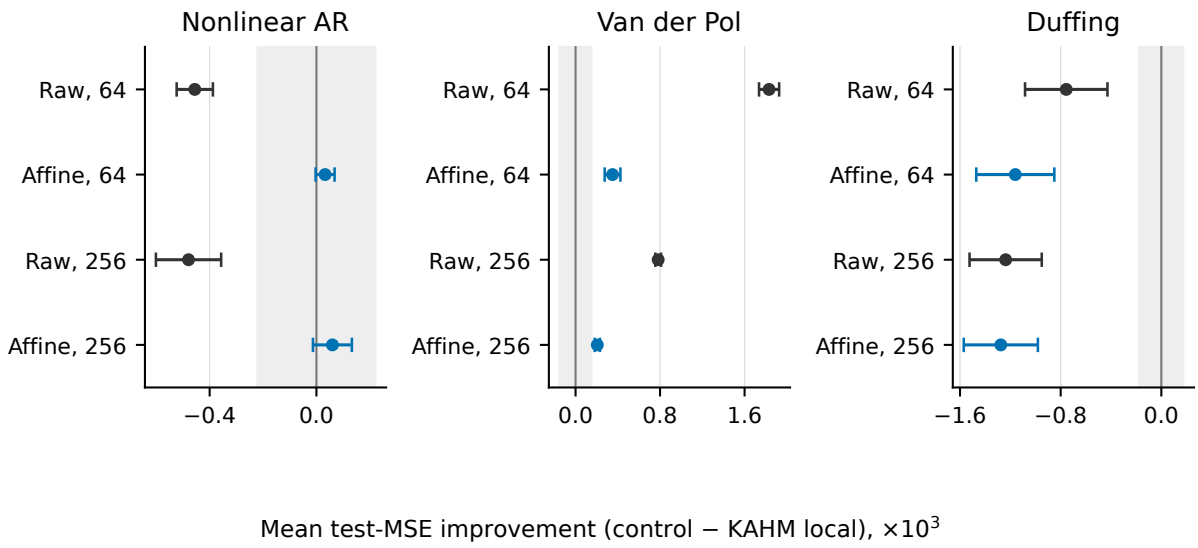


Figure 3: All 12 primary second-stage contrasts, each defined as comparator MSE minus KAHM-local MSE. Positive values favor KAHM. Intervals use the second-stage 95% Bonferroni family across 24 paired full data-and-fit repetitions. Gray bands mark the fixed ± practical margins from first-stage raw-RBF errors. This family is separate from the 29 first-stage contrasts.

6.4 Gaussian query underflow does not explain the local-feature advantage

The original query kernel can underflow when every Gaussian value in a query row becomes numerically zero. Because the kernel weights are normalized, multiplying a nonzero row by a common positive constant does not change the exact reconstruction. We therefore performed a post-hoc sensitivity analysis using the stable evaluation

$$\widetilde{k}_{cj}(z) = \exp\{\ell_{cj}(z) - \max_i \ell_{ci}(z)\}, \qquad \ell_{cj}(z) = -\frac{\Delta_{cj}(z)^\top \Sigma_c^{-1} \Delta_{cj}(z)}{2p_c}, \tag{17}$$

where $\Delta_{cj}(z) = P_c^\top (z - v_{cj})$, v_{cj} is reference j, p_c is the retained local rank, and Σ_c is the projected reference covariance. We compute ℓ_{cj} directly from this quadratic form, before exponentiation; taking logarithms of already underflowed kernel values would not repair an all-zero row. We keep the fitted references, basis, regularized matrix, and data unchanged. We refitted and validation-tuned the affected heads consistently under this query evaluation. This is a robustness analysis on reused second-stage Van der Pol data, not an additional independent confirmation.

Across 2,236,416 test query–local-model evaluations, the original implementation has 99,762 exactly zero weight normalizers; the stable evaluation has none. Despite this substantial numerical diagnostic, mean KAHM-local test MSE changes by less than 4×10^{-13} at either width. Its advantage over raw and affine local features remains positive in all 24 repetitions at both widths. Full-score features change more: their mean errors improve by approximately 0.37–0.39%, and some individual predictions change appreciably. We therefore do not claim that the numerical issue is irrelevant to every representation.

The practical implication is to stabilize the inner kernel evaluation and retain explicit diagnostics for near-zero or cancelling signed normalizers. Row scaling handles Gaussian underflow; it does not guarantee that signed weights can never cancel. The archived main results retain the original implementation so that the sensitivity comparison remains inspectable.

6.5 A natural-data check limits claims of general predictive superiority

The secondary Airfoil experiment favors the raw RBF predictor (Table 5). Its mean group-weighted MSE is 15.6898 dB^2, compared with 19.4726 for raw ridge, 20.90 for KAHM local-coordinate ridge, 31.86 for KAHM

Table 4: Post-hoc Van der Pol numerical sensitivity on the same 24 second-stage banks and samples. Stable query evaluation removes observed zero denominators while leaving local-coordinate mean MSE effectively unchanged. Δ_r is the original-minus-stable MSE difference within repetition r; wins count stable predictions beating the raw and affine local controls at the same width. These comparisons are outside the two primary interval families.

Encoding	M	Original MSE	Stable MSE	Max. $\lvert\Delta_r\rvert$	Wins raw; affine
Local	64	0.001026627	0.001026627	1.01×10^{-12}	24/24; 24/24
Local	256	0.000871969	0.000871969	1.69×10^{-11}	24/24; 24/24
Scores	64	0.001571085	0.001565219	8.74×10^{-5}	24/24; 4/24
Scores	256	0.001049909	0.001045801	3.53×10^{-5}	24/24; 17/24

Table 5: Secondary Airfoil static-regression results on one fixed group split, averaged over 24 fitting seeds. Units are dB^2. These seeds do not constitute independent natural-data replications, and the comparison is descriptive.

Representation	Ridge MSE	RBF_{64} MSE
Raw	19.4726	15.6898
Affine ϕ	34.1630	36.0476
KAHM ϕ	31.8586	34.3338
Affine local	22.7200	22.4050
KAHM local	20.9040	20.9313
KAHM scores	24.2168	21.7070

affinity ridge, and 34.16 for affine affinity ridge. The first-stage validation selector chooses raw RBF in all 24 fits. This is evidence against treating the KAHM construction as a default replacement for raw-input regression. Because all seeds share the same held-out operating-condition groups, their spread is not an uncertainty interval for generalization to arbitrary natural datasets or new group populations.

7 How the Findings Change the Model

7.1 Retain local position at the predictive interface

The first change is to expose local position instead of making every predictor rely on normalized reconstruction scores alone. Equation (7) is a compact implementation of that change. It retains the interpretation of affinities as local gates while giving the decoder coordinates that vary within a local region. The branch intervention and the eight positive local-coordinate contrasts support this change for the studied settings. The affine control shows that it is a general representation principle whose usefulness should not be attributed solely to KAHM reconstruction.

Retaining $\log Z$ is a different intervention: it recovers the score vector's absolute scale in exact arithmetic, but does not directly expose position. The two augmentations should therefore not be treated as interchangeable. Full scores and raw-input skips are useful controls when a score-based predictor underperforms, because they test different possible bottlenecks.

7.2 Select the representation together with its decoder

The second change is to evaluate a complete representation–decoder pair. The first-stage ridge ordering would miss the favorable Van der Pol KAHM-local result available to the nonlinear head. Conversely, the existence of that favorable pair does not justify choosing it on Duffing or AR. The implemented second-stage selector considers the eight combinations of four representations and two landmark widths. Each candidate's hyperparameters are selected by validation, and the candidate with the lowest validation MSE supplies the final prediction. Test data play no role in either decision. This protocol was specified before the fresh second-stage data were generated.

The selector chooses raw inputs with 64 landmarks in all 24 AR repetitions, KAHM local coordinates with 256 landmarks in all 24 Van der Pol repetitions, and affine local coordinates with 256 landmarks in 23 of the 24 Duffing repetitions; the remaining Duffing choice is raw input with 256 landmarks. Mean selector errors are approximately 0.004424, 0.000872, and 0.002792, respectively. These are secondary operational results for the declared candidate set, not an oracle comparison with all learning algorithms. Selection adds validation use and computational cost, and a new task can require a larger or differently designed validation set (Cawley & Talbot, 2010).

The companion implementation exposes `PredictiveGeometryRegressor` with training and validation inputs, optional validation groups, prediction, and state serialization. Its default candidate family reproduces this selector. A raw-input selection avoids geometric queries at prediction time. The reusable wrapper was assembled after the experiments, and its predictions were checked against the archived selection procedure; its existence does not constitute an additional experiment.

7.3 Use reconstruction as a diagnostic, not the sole selection objective

The third change is evaluative. A geometric model should report how its code treats target-relevant variation alongside input reconstruction. On a controlled distribution this can be done using a known conditional mean, a target intervention, or a distribution-preserving transformation. On an unknown task, a fitted target predictor can provide an exploratory diagnostic, but its errors and selection must be accounted for before using it to claim an information-loss certificate.

For KAHM development, a useful next objective is therefore to learn or select local gates for predictive performance while retaining direct local coordinates and a raw-input alternative. We have implemented validation selection among fixed candidates; we have not tested a new supervised training objective for the KAHM reconstruction bank. Such an objective is a consequence to investigate, rather than a result established here.

8 Scope, Limitations, and Research Direction

The positive evidence has a specific scope. It concerns the declared generators, training and validation sizes, observation maps, horizons, small local bank, feature scaling, decoder grids, and test distributions. The capacity follow-up guards against a favorable first-stage seed outcome and checks two finite budgets. It does not remove task-selection effects: Van der Pol was a promising setting observed in the first stage, and we explicitly used that observation to motivate the follow-up. The other two laws were retained to test the breadth of the finding.

The controlled comparison changes one reconstruction rule inside a matched construction. This supports component attribution, but it does not establish superiority over independently optimized affine local learning. Indeed, the locality-aware affine scoring control nearly closes the straight-branch gap. Local-coordinate augmentation also changes dimensionality and regularization geometry. Its gain cannot be assigned wholly to information recovery, and an equal nominal landmark count is not an exactly equal number of effective parameters. At 256 landmarks on Van der Pol, for example, the retained RBF-plus-linear dimensions are 268 for raw inputs, 298 for affine local coordinates, and approximately 272 on average for KAHM local coordinates after the eigenvalue cutoff. Table 12 reports the retained dimensions for all laws, representations, and landmark budgets.

The statistical analysis uses paired t approximations across independent synthetic repetitions. Bonferroni adjustment protects the declared family only to the extent that these marginal approximations are adequate. The intervals do not cover every exploratory analysis. Runtime measurements share intermediate computations across methods and are not fair standalone end-to-end speed comparisons. The numerical sensitivity exercise addresses one concrete implementation risk on reused Van der Pol data; it is not a complete conditioning analysis of signed kernel normalization.

The natural-data evidence is narrow and adverse for the presented KAHM variants. A broader claim about machine learning utility requires natural forecasting datasets, strong raw-input and local-model baselines, and

protocols that respect series or subject boundaries. We have not compared against a modern, broadly tuned forecasting suite or demonstrated state-of-the-art forecasting. The present contribution is an explanatory and reproducible study of a particular geometric model family.

Most importantly, there is not yet a validated diagnostic that predicts before fitting the downstream model when KAHM geometry will help on a new task. A credible next test would define such a diagnostic using training and validation data, freeze its prediction of relative benefit, and evaluate that prediction across held-out tasks. Candidate quantities include target-relevant changes along local directions and the sensitivity needed to decode the corresponding code, rather than reconstruction error alone. The test must include tasks with different observation maps, noise levels, sample sizes, and dynamics, and must retain cases where the prediction is wrong. This would convert a conditional empirical finding into transferable model-selection knowledge.

9 Conclusion

KAHM geometry can help prediction, but its benefit depends on what the predictive representation retains and which decoder uses it. In a controlled branch experiment, KAHM-derived scores preserve signed positional variation that a matched affine residual code nearly removes, despite worse reconstruction. Explicit local coordinates expose that variation for both geometries. In a separately designed fresh-sample forecasting study, KAHM local coordinates with a finite nonlinear decoder give a clear advantage on Van der Pol and lose on other declared laws. The resulting model design retains local position, allows decoder flexibility, and selects among geometric and raw-input alternatives using validation data. These findings support a precise development path for KAHMs: study and exploit target-relevant geometry, while testing the conditions of its advantage explicitly.

Code, Data, and Research Transparency

The accompanying `TMLR_Reproducibility_Supplement.zip` contains the source snapshot, fixed configurations and seed rules, compact historical results, statistical audit scripts, paper-asset generators, and commands for fresh experiment execution. Its reproduction guide distinguishes historical-result verification, a small execution check, and a complete regeneration. Fresh executions write separate output directories and provenance; they do not overwrite or retrospectively recreate the historical freezes. Generated model states and predictions support loss recomputation and selected-model replay. Synthetic data are generated from the specified laws. The included Airfoil data retain source, checksum, and license attribution.

The KAHM routines are pinned to the upstream commit

`d6d74d1953a4310b73065a556df816e5ab5e3ebc`.

The artifact records both source hashes and the original numerical environment. Seeds and package versions aid numerical reproduction but do not guarantee byte-identical floating-point results across hardware and BLAS implementations. Timing fields are not treated as deterministic scientific outcomes. Automated checks compare the regenerated predictive quantities and primary contrasts with the archived results within stated numerical tolerances.

Relationship to a related unpublished manuscript. Anonymous (2026) studies risk certificates for prescribed forecasting representation and decoder classes. The present paper asks a different empirical question about KAHM geometry, encoding, and finite fitted prediction. It uses shared upstream machinery and generator definitions, but the target-intervention experiments, fresh capacity follow-up, numerical sensitivity analysis, and validation-selected model constitute the study reported here. The ridge and RBF heads are not that manuscript's fixed sensitivity-budget classes, and no certificate from that manuscript is transferred to these fitted results. Anonymized copies are included as `Related_Manuscript.pdf` and `Related_Manuscript_Supplement.pdf` in the accompanying supplementary ZIP to support assessment of the relationship between the studies.

Use of AI tools. AI tools assisted with study implementation, numerical and statistical checks, literature retrieval, reproducibility packaging, and manuscript drafting. The executable artifact and recorded evidence are provided to support independent scrutiny of the claims. The authors are responsible for the manuscript's content and for checking the resulting claims.

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A Supporting Mathematical Statements

The statements in this appendix make the interpretation of the experiments precise. They are elementary supporting results; we do not claim a new KAHM approximation theorem. All conditional expectations below are understood up to almost-sure equality. For scalar square-integrable Y, write $m(X) = \mathbb{E}[Y \mid X]$ and $R_X^{\mathrm{Bayes}} = \mathbb{E}[(Y - m(X))^2]$. When a representation has been learned from independent training data, the statements apply conditionally on that fitted representation.

Proposition 1 (Information retained by a representation)**.** *For a measurable representation $U = \Gamma(X)$,*

$$R_U^{\mathrm{Bayes}} - R_X^{\mathrm{Bayes}} = \mathbb{E}[(m(X) - \mathbb{E}[m(X) \mid U])^2].$$

For any square-integrable prediction $f(U)$,

$$R(f \circ \Gamma) - R_X^{\mathrm{Bayes}} = \|m - f \circ \Gamma\|_{L^2}^2.$$

Proof. Write $Y - f(U) = (Y - m(X)) + (m(X) - f(U))$. The cross term has expectation zero after conditioning on X. Minimizing the second term over U-measurable predictions gives the conditional expectation $\mathbb{E}[m(X) \mid U]$. □

Proposition 2 (Restoring the outer score normalizer)**.** *Let $T_1, \ldots, T_C \in [0, 1]$, $\omega > 0$, and $\tau > 0$. Define*

$$\ell_c = \omega \log(1 - T_c + \tau), \qquad z = \log \sum_{j=1}^{C} \exp(\ell_j), \qquad \phi_c = \exp(\ell_c - z).$$

Then T and (ϕ, z) generate the same information in exact arithmetic, and

$$T_c = 1 + \tau - \exp\{(\log \phi_c + z)/\omega\}.$$

For multiple separately normalized blocks, one z is required per block.

Proof. Strict positivity of $1 - T_c + \tau$ implies $\phi_c > 0$. Hence $\log \phi_c + z = \ell_c$, which gives the inverse. Both directions are measurable. Applying this argument to each block gives the multiple-block statement. □

If scores are clipped before normalization, the inverse restores the clipped scores. It does not restore discarded reconstruction vectors or guarantee equal finite-head performance. In particular, the inverse is nonlinear, so affine function classes on T and on (ϕ, z) need not coincide. This proposition concerns the outer association normalizer, not the signed denominator in the KAHM reconstruction.

Proposition 3 (A sufficient condition for an injective local code)**.** *Fix a fitted bank with bases P_c and centers μ_c. Suppose its gates are measurable and strictly positive, and let A be the matrix obtained by vertically stacking the matrices $P_c^\top$. If A has full column rank d, then*

$$z \longmapsto \left[\phi(z), \{\phi_c(z) P_c^\top (z - \mu_c)\}_c\right]$$

is injective. In particular, this code and z generate the same information, and their Bayes risks agree for every square-integrable target.

Proof. The code contains both $\phi_c(z) > 0$ and its product with $P_c^\top(z - \mu_c)$. Division and addition of the fixed vector $P_c^\top \mu_c$ recover $P_c^\top z$ for every c. Stacking these vectors gives Az. Full column rank gives the measurable inverse $z = (A^\top A)^{-1} A^\top (Az)$. Conversely, the code is a measurable function of z, so the generated information and Bayes risks agree. □

The gates in Equation (6) are strictly positive in exact arithmetic because $\tau > 0$. The rank assumption is a sufficient condition, not a necessary one: when A is rank deficient, the gates themselves may retain additional information. It must be checked for each fitted bank. Very small gates and an ill-conditioned A can make the displayed inverse numerically unstable, and its existence does not make it accessible to every finite decoder or fitting procedure.

Proposition 4 (A score symmetry and its prediction consequence)**.** *Let Q be orthogonal on a preprocessed input space. Suppose that each group's reconstruction obeys $r_c(Qx) = Qr_c(x)$ and that Q preserves group identities. The radial–angular score in Section 3, its block affinities, and the block log normalizer are invariant under Q.*

For the branch law in Equation (10)*, let $Q(x_1, x_2) = (-x_1, x_2)$. Any representation $U = \Gamma(X)$ satisfying $\Gamma(QX) = \Gamma(X)$ almost surely has*

$$R_U^{\text{Bayes}} = \frac{1}{300} + \lambda + (1-\lambda)\mathbb{E}[(S - \mathbb{E}[S \mid U])^2].$$

Consequently $R_U^{\text{Bayes}} \geq 1/300 + \lambda$, with equality when U determines S.

Proof. Orthogonality preserves norms and inner products, including the zero-norm convention for the angle. Equivariance therefore preserves each radial and angular discrepancy, hence its combined score. Block normalization and its log normalizer are deterministic functions of those scores.

For the branch law, replacing T by $-T$ leaves the joint distribution of (S, T) unchanged and leaves U unchanged. Thus $\mathbb{E}[T \mid U] = 0$. Also S is even under this reflection, whereas T is odd. It follows that $\mathbb{E}[(S - \mathbb{E}[S \mid U])T] = 0$. Expanding the square in Proposition 1, using $\mathbb{E}[T^2] = 1/3$ and $\text{Var}(\epsilon) = 1/300$, yields the displayed identity. □

The first assertion also holds when Q permutes reconstructors within a group whose score is their minimum. Permuting distinct groups, however, can permute the affinity coordinates rather than leaving the representation unchanged. Nor does a raw-input reflection necessarily commute with a training-fitted scaler. Finite learned partitions and reference sets need not satisfy the proposition's assumptions. The measured near-invariance of one learned code is therefore a diagnostic, not a proof that its Bayes risk equals the exact-symmetry lower bound. Tiny nonzero differences could in principle be used by an unrestricted decoder.

Proposition 5 (Approximate symmetry under a decoder budget)**.** *Let Q be a measurable, distribution-preserving involution on the input space: $Q^2X = X$ almost surely and QX has the law of X. Let d_U be a measurable pseudometric on the representation space and g be κ-Lipschitz for this pseudometric. Assume $m, g \circ \Gamma \in L^2$ and*

$$\delta_\Gamma^2 = \mathbb{E}[d_U(\Gamma(X), \Gamma(QX))^2] < \infty.$$

With $m_o = (m - m \circ Q)/2$,

$$R(g \circ \Gamma) - R_X^{\text{Bayes}} \geq (\|m_o\|_{L^2} - \kappa\delta_\Gamma/2)_+^2.$$

For the branch law, $\|m_o\|_{L^2} = \sqrt{\lambda}$.

Proof. The operator $Vh = h \circ Q$ is an isometric involution on $L^2(P_X)$. The even and odd components $h_e = (h + Vh)/2$ and $h_o = (h - Vh)/2$ are orthogonal. In particular, with $f = g \circ \Gamma$,

$$\|m - f\|_{L^2}^2 = \|m_e - f_e\|_{L^2}^2 + \|m_o - f_o\|_{L^2}^2 \geq \|m_o - f_o\|_{L^2}^2.$$

The Lipschitz condition gives $|f_o(X)| \leq \kappa d_U(\Gamma(X), \Gamma(QX))/2$, so $\|f_o\|_{L^2} \leq \kappa\delta_\Gamma/2$. Apply the reverse triangle inequality and Proposition 1. For the branch law, $m_o(X) = \sqrt{3\lambda}\,T$ and $\mathbb{E}[T^2] = 1/3$. □

An empirical reflection distance is not automatically a population upper bound for δ_Γ. A valid numerical lower bound obtained from this proposition would need uncertainty control in that direction and a verified decoder Lipschitz constant for the same metric. Neither is asserted for the experiment's unconstrained diagnostic heads.

Proposition 6 (Reconstruction through the predictor's representation)**.** *Suppose a version of m is L-Lipschitz on a domain containing X and $h(U)$ almost surely, and $\mathbb{E}\|X - h(U)\|_2^2 \leq \varepsilon_{\text{rec}}^2$. Then*

$$R_U^{\text{Bayes}} - R_X^{\text{Bayes}} \leq L^2\varepsilon_{\text{rec}}^2.$$

If h is additionally L_h-Lipschitz from the declared representation metric and $\kappa \geq LL_h$, the same upper bound applies to the infimum excess risk over κ-Lipschitz decoders:

$$\inf_{\mathrm{Lip}(g)\leq\kappa} \{R(g \circ \Gamma) - R_X^{\mathrm{Bayes}}\} \leq L^2 \varepsilon_{\mathrm{rec}}^2.$$

If a particular decoder f instead satisfies $\|m \circ h - f\|_{L^2(P_U)} \leq \delta_{\mathrm{head}}$, then

$$R(f \circ \Gamma) - R_X^{\mathrm{Bayes}} \leq (L\varepsilon_{\mathrm{rec}} + \delta_{\mathrm{head}})^2.$$

Proof. Conditional expectation is the L^2 projection onto the U-measurable functions. The admissible prediction $m(h(U))$ gives

$$\mathbb{E}[(m(X) - \mathbb{E}[m(X) \mid U])^2] \leq \mathbb{E}[(m(X) - m(h(U)))^2] \leq L^2 \mathbb{E}\|X - h(U)\|_2^2.$$

For the constrained statement, $m \circ h$ has Lipschitz constant at most LL_h. For a particular f, the triangle inequality in L^2 gives $\|m - f \circ \Gamma\|_{L^2} \leq L\varepsilon_{\mathrm{rec}} + \delta_{\mathrm{head}}$. Use Proposition 1 in each case. □

The essential assumption is the existence of h as a function of the retained representation. A KAHM reconstruction evaluated on the raw query is not such a function merely because its residual is small. The domain assumption also matters: a conditional mean initially defined only on the support of X must have the specified Lipschitz version at $h(U)$.

Proposition 7 (Common scaling of an inner kernel row)**.** *Let k be a finite kernel row, let B be the fixed fitted matrix, and suppose $kB\mathbf{1} \neq 0$. For every real $a > 0$,*

$$\frac{akB}{akB\mathbf{1}} = \frac{kB}{kB\mathbf{1}}.$$

Thus evaluating a Gaussian query row by $\exp(\ell_j - \max_i \ell_i)$ preserves its normalized signed weights in exact arithmetic whenever the original denominator is nonzero.

Proof. The common factor cancels from numerator and denominator. For finite log-kernel values, the centered exponential row is the original row times $a = \exp(-\max_i \ell_i) > 0$ in real arithmetic. □

This statement neither rescales the training kernel nor permits replacing B. It does not address a genuinely zero signed denominator or prevent cancellation. The purpose of the implemented change is to avoid evaluating an entire query row as floating-point zero before normalization.

B Generators, Shared Components, and Decoders

B.1 Exact forecasting laws and target construction

The nonlinear AR generator starts with independent $u_0, u_1 \sim \mathrm{Unif}[-0.8, 0.8]$ and uses

$$u_{t+1} = \tanh\{1.25u_t - 0.65u_{t-1} + 0.25\sin(3u_t) + \eta_{t+1}\}, \qquad \eta_t \overset{\mathrm{iid}}{\sim} \mathrm{Unif}[-0.1, 0.1].$$

The first 80 stored values are discarded; the next 72 are the episode. Each episode uses an independent initial state and innovation sequence.

The Van der Pol and Duffing systems respectively use

$$\dot{x} = v, \qquad \dot{v} = (1 - x^2)v - x,$$
$$\dot{x} = v, \qquad \dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \theta).$$

Initial positions and velocities are independent $\mathrm{Unif}[-2, 2]$ draws for each episode, and $\theta \sim \mathrm{Unif}[0, 2\pi]$ is independently sampled for each episode. The phase draw is unused for Van der Pol but is retained in the

common generator's random-stream order. Both systems are integrated using float64 classical fourth-order Runge–Kutta with step 0.025, burn-in time 10, and observation interval 0.1. At the 72 observation times, the sensor is

$$u_t = 0.95\tanh(x_t) + \eta_t, \qquad \eta_t \overset{\text{iid}}{\sim} \text{Unif}[-0.02, 0.02].$$

Sensor noise is sampled before numerical integration in the frozen generator. These define numerical data laws. We do not treat the resulting samples as exact solutions of the continuous equations or assume stationarity after burn-in.

For all three laws, zero-based forecast origins are $11, 15, \ldots, 59$. At origin t, the input is $(u_{t-11}, \ldots, u_t)$ and the three targets are $(u_{t+1}, u_{t+4}, u_{t+12})$. There are 13 origins per episode. Training, validation, and test contain respectively 128, 128, and 512 independently generated episodes. They therefore contain 1,664, 1,664, and 6,656 rows, but rows within an episode are dependent.

The branch law is Equation (10). Training has 1,024 points for the primary study, with the first 256 points giving the secondary nested sample-size condition. Validation and test have 512 and 2,048 independent points. At each repetition, the same latent draws of S, T, ϵ generate both curvatures; within each curvature and training size, one fitted bank serves all three target mixtures. Latent labels never train the partitions. Since $T = X_1$ and $S = 2\{X_2 - c\cos(\pi X_1)\}$, the raw inputs determine the signal. Its variance is $(1-\lambda) + 3\lambda/3 = 1$, and the independent noise variance is $1/300$. The support bound $|Y| \le \sqrt{1-\lambda} + \sqrt{3\lambda} + 0.1 \le 2.1$ justifies the common clipping range.

The master random seed is 20261030. Separate phase identifiers and child streams determine pilot, first confirmation, natural fitting, and fresh capacity inputs and fits. The capacity phase identifier is 4. The archive contains the executable stream rules and the realized identifiers. Reusing a replicate label across phases does not reuse its observations. Common randomness within a repetition supports pairing; it does not create additional independent repetitions.

B.2 Training the common bank and defining the locality control

The input mean and population-standard-deviation convention ($1/n$ variance divisor) are fitted on all training rows. A standard deviation at most 10^{-12} is replaced by one. At most 1,600 training rows, sampled without replacement, are used for clustering and reference selection. Each of the three independent k-means block fits uses five initializations and at most 150 iterations. Its cluster counts are 2, 4, and 8. At most 24 reference rows per cluster are sampled without replacement and converted to float32 before the unchanged upstream autoencoder is called. The effective requested rank is $\min(2, d-1, n_c-1)$, subject to upstream numerical rank determination. The common reference mean and the resulting upstream principal basis define the affine control.

For completeness, the pinned local KAHM fit can be written as follows. Let $H \in \mathbb{R}^{d\times n_c}$ contain the float32 reference columns, and set $q_i = P_c^\top H_i$. The basis is obtained from the centered reference covariance with divisor $n_c - 1$, retaining eigenvalues above machine epsilon. The requested rank is reduced if a retained projected coordinate has range below 10^{-3}, until that condition is removed or rank one is reached. With effective rank r_c and unbiased projected covariance Σ_c, the training kernel is

$$K_{ij} = \exp\left\{-\frac{(q_i - q_j)^\top \Sigma_c^{-1}(q_i - q_j)}{2r_c}\right\}.$$

Writing $\nu = \|H\|_F^2/(dn_c)$, initialize $\rho_0 = 2.5\nu$ and iterate

$$B_j = (K + \rho_j I)^{-1}, \qquad \rho_{j+1} = 2\nu + \frac{\|H - H(KB_j)^\top\|_F^2}{dn_c}.$$

The routine stops when $|\rho_{j+1} - \rho_j| \le 10^{-3}$ or after 100 iterations, returning the last computed B_j. This regularizer is determined from reference inputs alone, independently of the supervised target. Query kernels use the same projected covariance metric and $1/(2r_c)$ Gaussian scale. A cluster with fewer than two usable references or no positive empirical-variance direction has the recorded center reconstruction fallback; other numerical failures are not silently replaced by another method.

The two reconstruction methods have the same input basis but not identical functional capacity. KAHM uses its nonlinear query weights to combine full-dimensional reference vectors; it need not remain inside the truncated affine PCA plane. A shared basis does not make its reconstruction an affine projection. The experiment isolates the reconstruction operator within the specified bank, not all possible advantages of independently optimized complete systems.

The alternative affine locality score is essential to interpreting the straight-branch comparison. In standardized input space, let

$$s_c = \max\left\{\frac{1}{n_c}\sum_{i\in c}\|z_i-\mu_c\|_2^2, 10^{-6}\right\},$$
$$v_c = \max\left\{\frac{1}{n_c}\sum_{i\in c}\|(I-P_cP_c^\top)(z_i-\mu_c)\|_2^2, 0.05s_c, 10^{-6}\right\}.$$

It uses

$$b_c(z) = \frac{\|z-r_c^{\mathrm{A}}(z)\|_2^2}{v_c} + 0.1\frac{\|z-\mu_c\|_2^2}{s_c}, \qquad \psi_c(z) = \frac{\exp(-b_c(z))}{\sum_{j \text{ in block}}\exp(-b_j(z))}.$$

This retains a center-distance term that distinguishes positions along an otherwise exactly reconstructed line. It changes the scoring rule and is a secondary control, not part of the primary reconstruction substitution. A radial-basis locality control similarly normalizes $\exp\{-\|z-\mu_c\|_2^2/(2\,\mathrm{median}_j\, s_j)\}$ within a block.

There are 14 stored affinity coordinates and three block-sum constraints, giving an affine-dimension upper bound of 11 for the concatenated affinities. Appending the three log normalizers gives 17 stored coordinates. Local augmentation gives 28 coordinates for the two-dimensional branches and 42 for the 12-dimensional forecasting histories. The added coordinates are $\phi_cP_c^\top(z-\mu_c)$, computed from the original query in the shared basis; full reconstruction vectors are not passed to these heads.

For reconstruction diagnostics, each query's squared reconstruction distance is divided by input dimension. The best-model diagnostic takes the minimum across local models in each block; the weighted diagnostic takes the sum with that reconstruction's own affinity weights. Both then average across queries, blocks, and repetitions. Thus the weighted summary is an affinity-weighted average of individual errors, not the reconstruction error of the affinity-weighted vector. These descriptive summaries are not additional primary forecast contrasts.

B.3 Ridge and finite nonlinear heads

Each input feature column is standardized using training data alone. Standard deviations below 10^{-8} are replaced by one. The ridge fit minimizes mean squared training error plus $\alpha\|W\|_F^2$, with an unpenalized intercept. The fixed penalty grid is

$$\alpha \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}.$$

Validation chooses the penalty. Forecast horizons share that choice; the three branch targets choose separately.

The nonlinear head first selects M training rows without replacement as landmarks in the standardized input-feature space. For a fixed budget, the common seed selects the same row indices across the representations, though their feature values differ. Let b^2 be the median strictly positive squared landmark distance, excluding distances at most 10^{-14}; if no such distance exists, use $b^2 = 1$. For bandwidth multiplier $a \in \{0.25, 1, 4\}$, the squared bandwidth is $\max(b^2a^2, 10^{-12})$. Form the Gaussian landmark matrix K_{MM}, symmetrize it, and retain eigenvalues exceeding $\max(10^{-10}\lambda_{\max}, 10^{-12})$. If $K_{MM} = V\Lambda V^\top$ on the retained subspace, the nonlinear coordinates are $K_{xM}V\Lambda^{-1/2}$. Concatenate these coordinates with the standardized linear input features; standardize the full resulting basis on training data and fit the same ridge objective. Validation selects a bandwidth–penalty pair using the fixed nested iteration order, so ties are deterministic. The first stage uses $M = 64$; the fresh follow-up uses both 64 and 256.

Because basis columns are standardized and assigned an ordinary Euclidean ridge penalty, this is a finite nonlinear diagnostic with a linear component. It is not an exact kernel-ridge estimator with its standard RKHS

norm. Effective dimension can be smaller than input dimension plus M after the eigenvalue cutoff. Increasing M can change the sampled landmark set, median-distance bandwidth, retained rank, and validation-selected hyperparameters. It does not isolate only a dimension parameter while holding the realized basis fixed.

Predictions are clipped to the known target range for both validation and test evaluation, but the ridge objective itself is the un-clipped training least-squares objective. The selected coefficients remain fitted on training data; validation observations are not used in a final combined fit. The secondary capacity selector chooses the lowest validation MSE among the eight complete representation–budget pipelines. Its fixed tie order is ascending budget followed by the configured feature order. The test split chooses neither hyperparameters nor the complete pipeline.

B.4 Airfoil grouping and the scope of its repetitions

The Airfoil snapshot contains 1,503 rows with five covariates and sound pressure as the target. Exact joint equality of the four non-frequency covariates defines 106 operating-setting groups. A fixed permutation with seed 2026103007 allocates 64, 21, and 21 groups to training, validation, and test, giving 905, 295, and 303 rows. All frequency rows within a setting stay together. The training objective weights rows equally; validation and test average within a group before weighting groups equally. The outcomes are reported in squared decibels.

The 24 seeds repeat fitting on this same observed cohort and split. They characterize fitting variability conditional on those data. No primary synthetic-experiment interval is applied to Airfoil, and the 24 fits are not treated as 24 independent draws of natural tasks, test cohorts, or physical operating-condition populations. A population claim would require a justified sampling model or independent cohorts.

C Estimands, Multiplicity, and Descriptive Diagnostics

C.1 The unit of paired inference

For forecast method A and repetition r, the reported loss is

$$\widehat{L}_{r,A} = \frac{1}{512}\sum_{j=1}^{512}\frac{1}{13\cdot 3}\sum_{o=1}^{13}\sum_{h\in\{1,4,12\}}(\widehat{y}_{r,A,j,o,h} - y_{r,j,o,h})^2.$$

For branches it is the mean noisy-target squared error over 2,048 test points. The conditional-mean target loss is a separate secondary diagnostic. Each method and target at a given repetition shares the declared test draws, yielding a paired contrast. The independent units for the primary intervals are the 24 complete train–validation–test repetitions. Overlapping windows, horizons, reference groups, candidate fits, and three target mixtures are not extra replication units.

For each primary paired contrast D_r, let $\overline{D}$ be its sample mean and s_D its unbiased sample standard deviation over 24 repetitions. For family size F, the interval is

$$\overline{D} \;\pm\; t_{23,\,1-0.05/(2F)}\,\frac{s_D}{\sqrt{24}}.$$

The critical values are approximately 3.545808 for $F = 29$ and 3.180663 for $F = 12$. Under independent normal paired differences, each marginal Student interval has its exact nominal coverage, and the Bonferroni union bound gives family coverage at least 95%. Under other laws, the Student coverage is approximate. Twenty-four repetitions do not guarantee an adequate normal approximation. The results are not distribution-free certificates. Dependence between contrasts is allowed by Bonferroni; independence of windows is never used.

The synthetic estimand includes training and validation randomness, as well as finite test-sample variation, under each declared numerical generator. It is not an average over an independently sampled population of forecasting tasks. The sample size of 24 was fixed as a resource decision, not justified by an unreported prospective power calculation.

C.2 Comparison families and practical margins

The first-stage family contains the 27 G, N, L contrasts over nine settings and the two interactions $I_c = G(c, \lambda = 0) - G(c, \lambda = 1)$. A positive interaction would favor a larger reconstruction advantage for the branch-identity target. Negative I_c is the opposite direction of this hypothesis, not a literal statement that one fitted method is uniformly harmful. The cell margin is 5% of the mean raw-ridge MSE from two pilot repetitions. Each interaction margin is the mean of its two endpoint cell margins. All pilot outcomes are excluded from confirmation estimates. The margins are recorded design thresholds, not externally validated application tolerances; their scale estimate has only two pilot observations.

The fresh capacity family contains 12 differences: raw minus KAHM local and affine local minus KAHM local, for three systems and two budgets. Each law's margin is 5% of its first-stage mean raw-RBF MSE, using the complete 24 first-stage confirmation repetitions. The margins, candidate list, and 24 fresh replication streams were frozen before phase-4 data were generated. Conditional on the completed first stage, they are fixed for this follow-up. The follow-up is explicitly data-motivated; it is not retroactively part of the initial 29-comparison plan. Its separate 95% family does not imply joint 95% coverage with the first family.

We call a difference a practical benefit at the declared threshold only when its simultaneous lower endpoint exceeds its fixed margin. An interval strictly above zero but not above the margin establishes a positive difference under the interval assumptions without resolving that practical threshold. A negative interval indicates an adverse difference for the candidate. An interval containing zero leaves the direction unresolved. An interval entirely within the fixed band $[-\delta, \delta]$ can support that band; failure to exclude zero alone cannot. Percent changes in the paper are descriptive ratios of means, without an asserted ratio confidence interval.

The capacity family has no primary interaction testing whether the advantage changes with landmark budget. A positive interval at one budget and an unresolved interval at another would not prove a budget interaction. All declared cells, including adverse and unresolved results, are retained. There were no omitted failed prescribed repetitions in the completed archives.

C.3 Reflection, reconstruction, and numerical checks

Reflection is applied to the raw branch input before the frozen scaler and geometry. For a p-coordinate feature vector, the stored-coordinate diagnostic is $\{\mathbb{E}_n\|\Gamma(X) - \Gamma(QX)\|_2^2/p\}^{1/2}$. The standardized diagnostic first divides each coordinate by its training-only standard deviation, replacing values at most 10^{-12} by one. This threshold differs from the fitted heads' 10^{-8} rule. Thus the reported per-coordinate RMS, its standardization, and its finite-sample approximation are not silently identified with the population metric or a known Lipschitz budget in Proposition 5. Values in Table 9 are averages of per-repetition RMS diagnostics, not a single pooled RMS.

The post-hoc numerical check retains all 24 phase-4 Van der Pol banks and all train/validation/test observations. It changes only query-row Gaussian evaluation as in Proposition 7, before multiplication by the unchanged fitted B. The training kernel, regularizer, principal basis, reference values, covariance metric, partition, and scaler are unchanged. Heads are fitted on the stabilized training features and retuned on the same validation split. The 96 outcomes cover two encodings, two budgets, and 24 repetitions. They are descriptive reuse of phase-4 data; no third confirmatory family or new confidence claim is attached.

The independently checked test count of 99,762 zero normalizers refers to query–local-model pairs among 2,236,416 evaluations, not distinct test observations. Of those pairs, 99,687 already had all Gaussian entries equal to floating-point zero; the remaining 75 zeros arose in later arithmetic. The stabilized count was zero. Local-feature losses were effectively unchanged, while full-score losses and some predictions changed. This addresses that particular underflow pathway as an explanation of the local-feature benefit; it is not a proof of numerical robustness of every KAHM encoding or of signed normalization in general.

C.4 Retrospective influence and coordinate-recovery diagnostics

These descriptive checks were added after the original study. They reuse the declared paired repetitions and do not change the primary estimates, margins, or comparison families. For each of the 41 primary contrasts,

we compute bias-corrected sample skewness and delete each repetition once. Each deletion recomputes the Student interval with 23 remaining repetitions, 22 degrees of freedom, and the original family size (29 or 12). The range of deletion endpoints is an influence diagnostic, not a confidence interval or an additional simultaneous-coverage claim. Table 6 shows the four Van der Pol comparisons; the complete descriptive results accompany the executable artifact. All four Van der Pol comparisons retain lower endpoints above the fixed practical margin after any one-repetition deletion. This does not establish a distribution-free mean guarantee or remove the normal-approximation qualification of the primary analysis.

Table 6: Retrospective Van der Pol influence diagnostics. Lower endpoints are multiplied by 10^3; the fixed margin on this scale is 0.148907. Skewness is the bias-corrected Fisher–Pearson sample coefficient. The minimum deletion lower endpoint is the smallest over all 24 one-repetition deletions. The final column counts individual paired differences exceeding the margin; it is not a mean-effect confidence statement.

Control	M	Skewness	Original lower	Min. deletion lower	$D_r > \delta$
Raw	64	-0.07	1.7369	1.7278	24/24
Affine local	64	2.69	0.2755	0.2685	24/24
Raw	256	0.01	0.7550	0.7524	24/24
Affine local	256	0.33	0.1810	0.1786	23/24

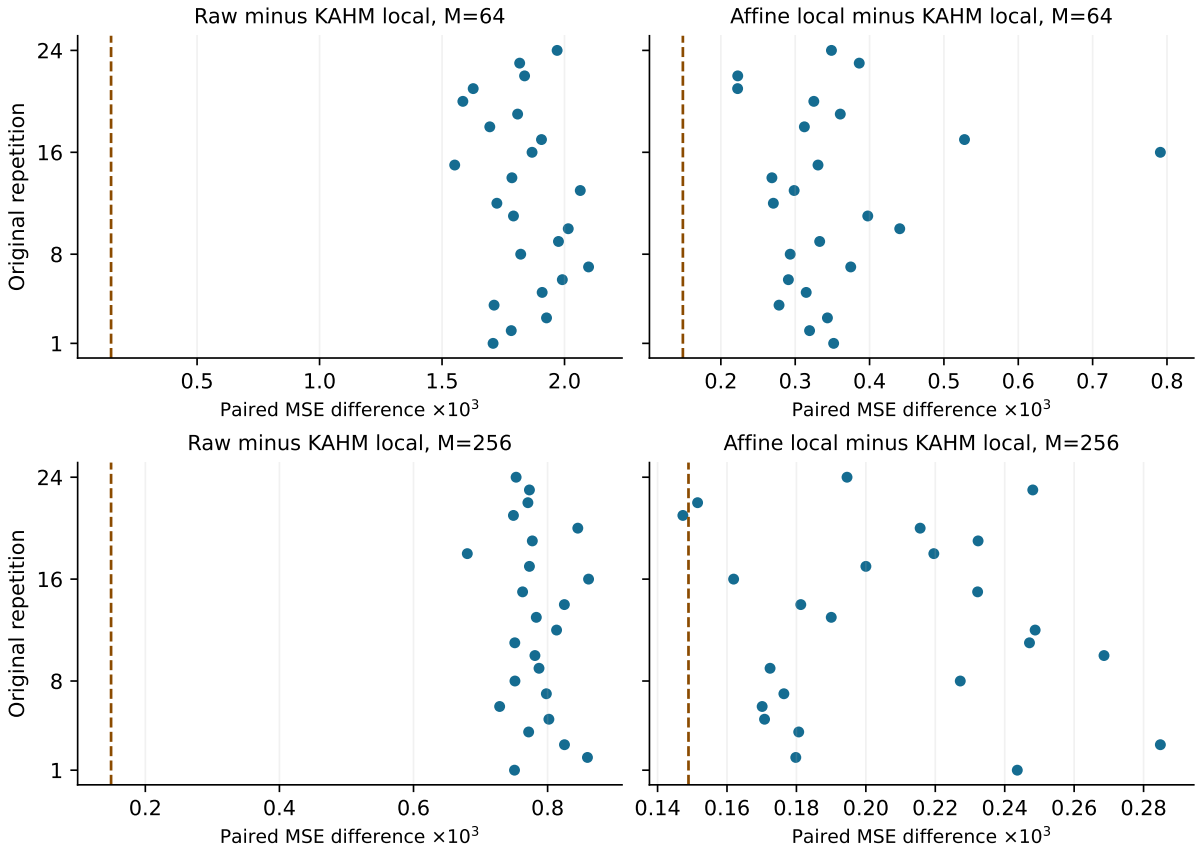


Figure 4: The 24 original paired differences for each Van der Pol capacity comparison, in units of 10^{-3} MSE. The dashed line is the frozen practical margin. No resampling, refitting, or new primary analysis is performed.

We also inspect the 120 first-stage primary banks and 72 capacity banks from a completed computational reproduction using the published seeds. No bank or head is refitted for these diagnostics. Replayed test inputs, targets, and groups are checked against their recorded hashes before the original feature map is evaluated. Let

A vertically stack the $P_c^\top$ matrices. Proposition 3 shows that positive gates and full column rank of A suffice to recover the scaled input from the gated local coordinates and gates: division by ϕ_c recovers $P_c^\top(z-\mu_c)$, after which one solves for z. A deficient stack does not establish information loss in the full code, because the gates can carry additional information. Numerical ranks use the threshold $\sigma_1 \max\{\mathrm{rows}(A), d\}\epsilon_{64}$; they are finite-precision diagnostics, not exact rank proofs. Small singular values and small gates can make coordinate recovery sensitive to perturbations. Test-sample gate minima are not population lower bounds, and these computations do not guarantee a well-conditioned inverse or easy decoding by any particular fitted head.

Table 7: Retrospective diagnostics over 24 stored banks per row. Rank ranges concern the stacked local bases in input dimension d. The smallest singular value and gate entries are minima over the inspected banks (and, for gates, their original test observations); κ_2 is the largest stack condition number. A dash means that not every stack in that row has full numerical column rank. KAHM and affine representations share the basis stack. Computational reproduction supplies no new independent statistical replications.

Phase / setting	Rank / d	Min. σ_d	Max. κ_2	Min. ϕ^{K}	Min. ϕ^{A}
First: Straight branch	1/2	0	—	3.36×10^{-25}	2.56×10^{-20}
First: Curved branch	2/2	2.19×10^{0}	1.39×10^{0}	3.55×10^{-28}	8.97×10^{-29}
First: Nonlinear AR	12/12	6.20×10^{-2}	4.90×10^{1}	2.21×10^{-35}	2.36×10^{-34}
First: Van der Pol	12/12	1.60×10^{-2}	2.20×10^{2}	2.84×10^{-45}	9.43×10^{-20}
First: Duffing	12/12	1.33×10^{-2}	2.80×10^{2}	3.92×10^{-43}	3.11×10^{-7}
Capacity: Nonlinear AR	12/12	7.31×10^{-2}	4.19×10^{1}	9.97×10^{-37}	7.75×10^{-37}
Capacity: Van der Pol	12/12	1.60×10^{-2}	2.19×10^{2}	6.21×10^{-44}	1.56×10^{-16}
Capacity: Duffing	12/12	1.32×10^{-2}	2.84×10^{2}	5.38×10^{-43}	1.86×10^{-8}

D Additional Source-Derived Tables

Tables 8 and 9 give descriptive input-geometry summaries. The two complete primary comparison families follow. Their values are generated from the archived outcomes, with each difference, standard error, interval, and frozen margin independently recomputed by the paper-asset script before it writes these tables.

Table 8: Held-out reconstruction errors per standardized input coordinate, averaged over the three blocks and 24 full repetitions. "Best" takes the best local reconstruction separately in each block; "weighted" averages the individual squared errors with that method's affinities. The affine value is lower in all 24 repetitions of each of these five input settings, for both summaries. These are descriptive diagnostics, not the supervised primary endpoint.

Setting	KAHM best	Affine best	KAHM weighted	Affine weighted
Branch $c = 0$	3.276×10^{-3}	2.156×10^{-16}	4.006×10^{-3}	1.548×10^{-8}
Branch $c = 1$	3.583×10^{-2}	2.932×10^{-2}	3.793×10^{-2}	3.238×10^{-2}
Nonlinear AR	1.166×10^{-2}	1.084×10^{-2}	1.305×10^{-2}	1.214×10^{-2}
Van der Pol	4.353×10^{-3}	1.683×10^{-3}	4.495×10^{-3}	1.916×10^{-3}
Duffing	4.877×10^{-3}	5.613×10^{-4}	5.239×10^{-3}	1.226×10^{-3}

Table 9: Branch reflection differences at training size 1,024, averaged over 24 banks. The statistic is a per-coordinate RMS, with and without training-only coordinate scaling. The raw reflection precedes the frozen pipeline. These values neither establish exact information loss for the learned code nor certify a decoder-specific population risk bound.

c	Encoding	Stored-coordinate RMS	Standardized RMS
0	KAHM ϕ	4.894×10^{-1}	1.402×10^{0}
0	Affine ϕ	1.451×10^{-9}	7.533×10^{-9}
0	KAHM local	5.289×10^{-1}	1.418×10^{0}
0	Affine local	4.993×10^{-1}	1.236×10^{0}
1	KAHM ϕ	5.380×10^{-1}	1.427×10^{0}
1	Affine ϕ	5.321×10^{-1}	1.471×10^{0}
1	KAHM local	4.854×10^{-1}	1.442×10^{0}
1	Affine local	5.522×10^{-1}	1.473×10^{0}

Table 10: Complete representation means in the fresh capacity study, with MSE multiplied by 10^3. The full-score KAHM representation is a prespecified secondary control. These point estimates do not add hypotheses to the 12-comparison primary family.

System	M	Raw	Affine local	KAHM local	KAHM scores
Nonlinear AR	64	4.4238	4.9120	4.8795	4.7984
Nonlinear AR	256	4.6958	5.2347	5.1749	4.8256
Van der Pol	64	2.8586	1.3767	1.0266	1.5711
Van der Pol	256	1.6541	1.0780	0.8720	1.0499
Duffing	64	3.4597	3.0549	4.2154	4.7005
Duffing	256	2.8364	2.7981	4.0736	3.3808

D.1 Selected pipelines and realized dimensions

The validation-selected pipeline and the retained nonlinear basis are distinct summaries: the former describes the operational model choice, whereas the latter records the realized dimensions behind the nominal landmark budgets.

Table 11: Secondary validation-only selection among the eight complete capacity-study pipelines. Mean test MSE is multiplied by 10^3. Counts concern independent complete repetitions under each generator, not a new population of forecasting tasks.

System	Selector MSE	Validation selections
Nonlinear AR	4.4238	Raw, 64: 24/24
Van der Pol	0.8720	KAHM local, 256: 24/24
Duffing	2.7923	Affine local, 256: 23/24; Raw, 256: 1/24

Table 12: Retained RBF-plus-linear dimensions after the Nyström eigenvalue cutoff in the fresh capacity study. Means and ranges cover the 24 repetitions in each cell. The same landmark count M does not imply the same realized dimension or function class.

System	M	Representation	Mean dimension	Range
Nonlinear AR	64	Raw	74.88	61–76
Nonlinear AR	64	Affine local	106.00	106–106
Nonlinear AR	64	KAHM local	106.00	106–106
Nonlinear AR	64	KAHM scores	76.71	62–78
Nonlinear AR	256	Raw	189.04	78–268
Nonlinear AR	256	Affine local	297.62	296–298
Nonlinear AR	256	KAHM local	297.50	293–298
Nonlinear AR	256	KAHM scores	148.88	77–219
Van der Pol	64	Raw	76.00	76–76
Van der Pol	64	Affine local	106.00	106–106
Van der Pol	64	KAHM local	106.00	106–106
Van der Pol	64	KAHM scores	78.00	78–78
Van der Pol	256	Raw	268.00	268–268
Van der Pol	256	Affine local	298.00	298–298
Van der Pol	256	KAHM local	272.38	261–285
Van der Pol	256	KAHM scores	231.88	215–251
Duffing	64	Raw	76.00	76–76
Duffing	64	Affine local	106.00	106–106
Duffing	64	KAHM local	105.46	98–106
Duffing	64	KAHM scores	77.96	77–78
Duffing	256	Raw	268.00	268–268
Duffing	256	Affine local	296.92	272–298
Duffing	256	KAHM local	240.38	176–298
Duffing	256	KAHM scores	214.33	164–270

Table 13: Complete first-stage family (29 contrasts), in original MSE units. Intervals use the two-sided 95% Bonferroni paired-replicate Student procedure with 23 degrees of freedom. Under paired normality, each t interval is exact and Bonferroni family coverage is at least 95%; otherwise coverage is approximate. The two stages have separate families.

Setting / budget	Contrast	$\overline{D}$	Simultaneous interval	δ	Assessment
$c = 0, \lambda = 0$	G	-0.000013	$[-0.000031, 0.000005]$	0.000168	unresolved †
$c = 0, \lambda = 0$	N	-0.000008	$[-0.000016, -1.85 \times 10^{-7}]$	0.000168	negative †
$c = 0, \lambda = 0$	L	-0.000031	$[-0.000043, -0.000020]$	0.000168	negative †
$c = 0, \lambda = 0.5$	G	0.489453	[0.481962, 0.496944]	0.000168	practical +
$c = 0, \lambda = 0.5$	N	0.000944	[0.000088, 0.001799]	0.000168	positive
$c = 0, \lambda = 0.5$	L	0.010616	[0.010033, 0.011200]	0.000168	practical +
$c = 0, \lambda = 1$	G	0.978026	[0.962552, 0.993500]	0.000168	practical +
$c = 0, \lambda = 1$	N	0.001938	[0.000255, 0.003621]	0.000168	practical +
$c = 0, \lambda = 1$	L	0.021287	[0.020199, 0.022376]	0.000168	practical +
$c = 1, \lambda = 0$	G	0.001761	$[-0.071260, 0.074781]$	0.033542	unresolved
$c = 1, \lambda = 0$	N	0.069128	[0.028663, 0.109593]	0.033542	positive
$c = 1, \lambda = 0$	L	0.263144	[0.217187, 0.309101]	0.033542	practical +
$c = 1, \lambda = 0.5$	G	-0.004395	$[-0.043672, 0.034882]$	0.016863	unresolved
$c = 1, \lambda = 0.5$	N	0.040663	[0.019789, 0.061538]	0.016863	practical +
$c = 1, \lambda = 0.5$	L	0.154710	[0.122867, 0.186552]	0.016863	practical +
$c = 1, \lambda = 1$	G	0.011324	[0.000882, 0.021767]	0.000168	practical +
$c = 1, \lambda = 1$	N	0.012111	[0.005647, 0.018574]	0.000168	practical +
$c = 1, \lambda = 1$	L	0.046854	[0.040400, 0.053307]	0.000168	practical +
Nonlinear AR	G	-1.89×10^{-7}	$[-0.000050, 0.000049]$	0.000253	unresolved †
Nonlinear AR	N	-0.000016	$[-0.000062, 0.000029]$	0.000253	unresolved †
Nonlinear AR	L	0.000433	[0.000368, 0.000498]	0.000253	practical +
Van der Pol	G	-0.049419	$[-0.060532, -0.038307]$	0.002036	negative
Van der Pol	N	0.013443	[0.005084, 0.021803]	0.002036	practical +
Van der Pol	L	0.096633	[0.092436, 0.100830]	0.002036	practical +
Duffing	G	-0.030948	$[-0.035414, -0.026482]$	0.000338	negative
Duffing	N	0.005558	[0.001721, 0.009395]	0.000338	practical +
Duffing	L	0.046168	[0.041729, 0.050608]	0.000338	practical +
$c = 0$ interaction	I_0	-0.978039	$[-0.993512, -0.962566]$	0.000168	opposite sign
$c = 1$ interaction	I_1	-0.009564	$[-0.082839, 0.063712]$	0.016855	unresolved

†: the entire interval lies within the fixed practical band $[-\delta, \delta]$. An unresolved sign alone does not establish equivalence.

Table 14: Complete fresh capacity family (12 contrasts). All numerical loss quantities are multiplied by 10^3. Intervals use the two-sided 95% Bonferroni paired-replicate Student procedure with 23 degrees of freedom. Under paired normality, each t interval is exact and Bonferroni family coverage is at least 95%; otherwise coverage is approximate. The two stages have separate families.

Setting / budget	Contrast	$\overline{D}$	Simultaneous interval	δ	Assessment
Nonlinear AR / 64	Raw–K	-0.4556	$[-0.5235, -0.3878]$	0.2201	negative
Nonlinear AR / 64	Affine–K	0.0325	$[-0.0031, 0.0680]$	0.2201	unresolved †
Nonlinear AR / 256	Raw–K	-0.4791	$[-0.6013, -0.3569]$	0.2201	negative
Nonlinear AR / 256	Affine–K	0.0598	$[-0.0131, 0.1327]$	0.2201	unresolved †
Van der Pol / 64	Raw–K	1.8320	$[1.7369, 1.9271]$	0.1489	practical +
Van der Pol / 64	Affine–K	0.3501	$[0.2755, 0.4247]$	0.1489	practical +
Van der Pol / 256	Raw–K	0.7822	$[0.7550, 0.8094]$	0.1489	practical +
Van der Pol / 256	Affine–K	0.2060	$[0.1810, 0.2311]$	0.1489	practical +
Duffing / 64	Raw–K	-0.7557	$[-1.0831, -0.4282]$	0.1727	negative
Duffing / 64	Affine–K	-1.1605	$[-1.4707, -0.8503]$	0.1727	negative
Duffing / 256	Raw–K	-1.2372	$[-1.5242, -0.9503]$	0.1727	negative
Duffing / 256	Affine–K	-1.2756	$[-1.5702, -0.9810]$	0.1727	negative

†: the entire interval lies within the fixed practical band $[-\delta, \delta]$. An unresolved sign alone does not establish equivalence.